AMAL JYOTHI
COLLEGE OF ENGINEERING
AUTONOMOUS
KANJIRAPPALLY

COMBINED ASSIGNMENT

23INMCA303 – Use Interface Design, 23INMCA305 – Introduction to RDBMS and SQL

Submitted To

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ABSTRACT

Background

The Kerala State Electricity Board (KSEB) Quick Pay portal plays a crucial role in enabling millions of consumers across Kerala to pay bills, lodge complaints, and request new connections. Despite being an essential digital service, the existing portal suffers from multiple usability limitations. Issues such as complex navigation, redundant authentication steps, unclear bill visibility, and scattered access to receipts have reduced user satisfaction and efficiency.

Problem Statement

The current KSEB Quick Pay portal has an outdated design and fails to adhere to key usability standards. Consumers face difficulties in recognizing service pathways, encounter vague error prompts during login, and often repeat actions unnecessarily. These design flaws create barriers to digital adoption and hinder smooth service delivery for the public.

Methodology

This study employed Nielsen's Ten Usability Heuristics as the evaluation framework for identifying problems and redesigning the interface. A systematic heuristic evaluation highlighted pain points such as poor visibility of system status, reliance on recall, lack of error prevention, and scattered access to critical services. A redesigned prototype was developed using Figma, with improvements aligned to heuristic principles including *recognition rather than recall, error prevention, aesthetic and minimalist design, and consistency and standards*.

Key Modules in Redesigned Portal

1. Quick Pay Module – Simplified form with clear instructions, placeholders, and contextual hints.
2. Bill Management Module – Structured dashboard displaying due amount, billing period, and payment status.
3. Complaint Management Module – Dropdown-based categorized complaints with guided text fields.

4. New Connection Module – A unified, structured form for all new connection requests.
5. Receipts and Downloads Module – Centralized access to receipts and transaction history.
6. User Authentication Module – Streamlined login with reduced redundancy and error-preventive guidance.

User Flow

- Homepage Access – Users encounter structured cards with icons for key services.
- Authentication – Simplified login or direct quick pay access.
- Bill Viewing & Payment – Pending bills visible immediately; error-free guided payment.
- Complaint Submission – Categorized and structured forms for precision.
- New Connection Request – Single structured form replaces multiple confusing links.
- Receipt Access – Dedicated downloads section for easy retrieval.

Existing Component Evaluation and Justification

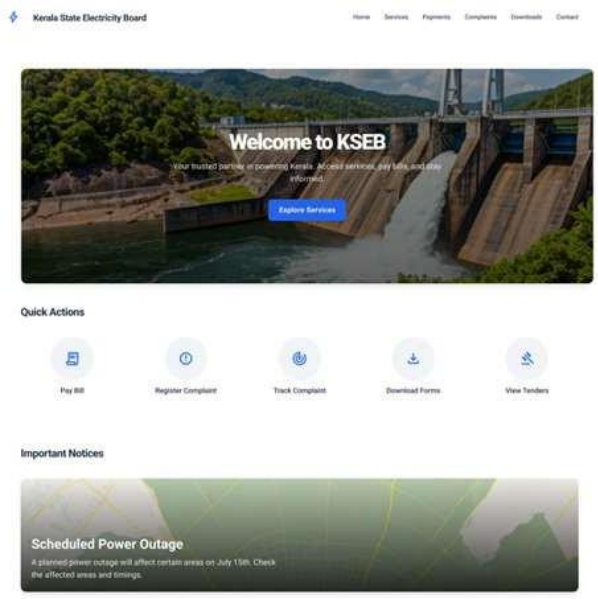
The original portal relied on plain text links, requiring users to recall navigation paths. Redesign introduced icon-based cards, addressing *recognition rather than recall*. The login process, previously burdened by CAPTCHA repetition, was simplified to ensure *error prevention* and *match with real-world standards*. Bill visibility was improved to enhance *system status visibility*. Complaint submission was redesigned for *user control and freedom*, while receipts were consolidated to support *error recovery*.

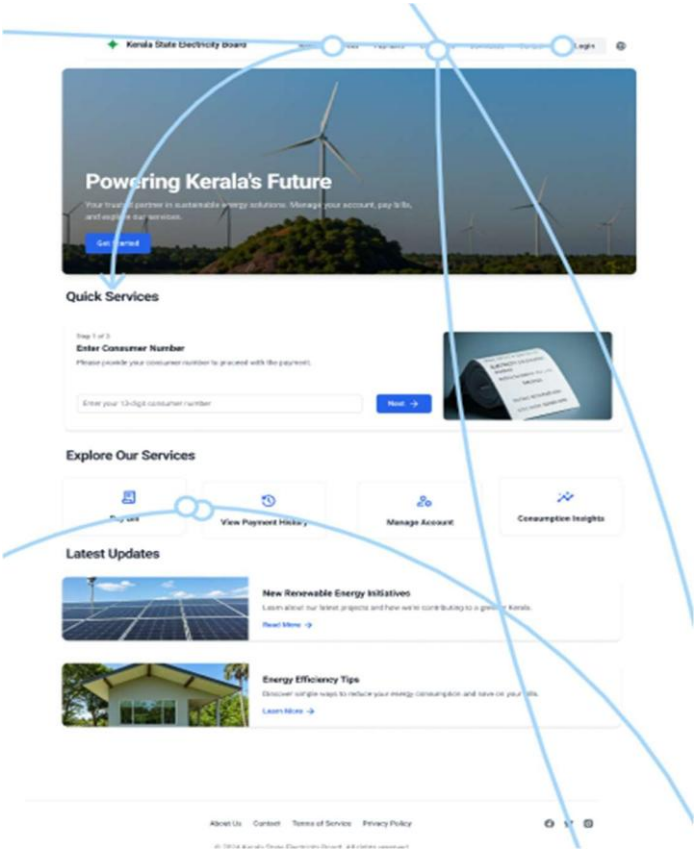
Conclusion

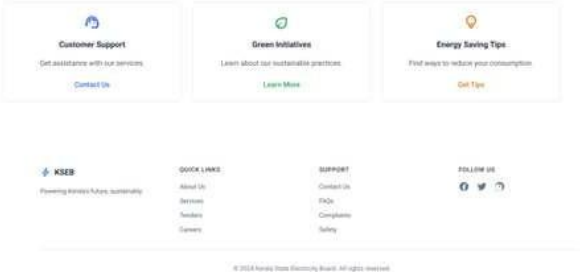
The heuristic-driven redesign of the KSEB Quick Pay portal ensures a modernized, user-centric interface. It enhances accessibility, reduces cognitive load, and improves service efficiency. By systematically applying usability heuristics, the redesigned portal addresses existing gaps, resulting in faster transactions, error-free navigation, and a more inclusive experience for all consumers.

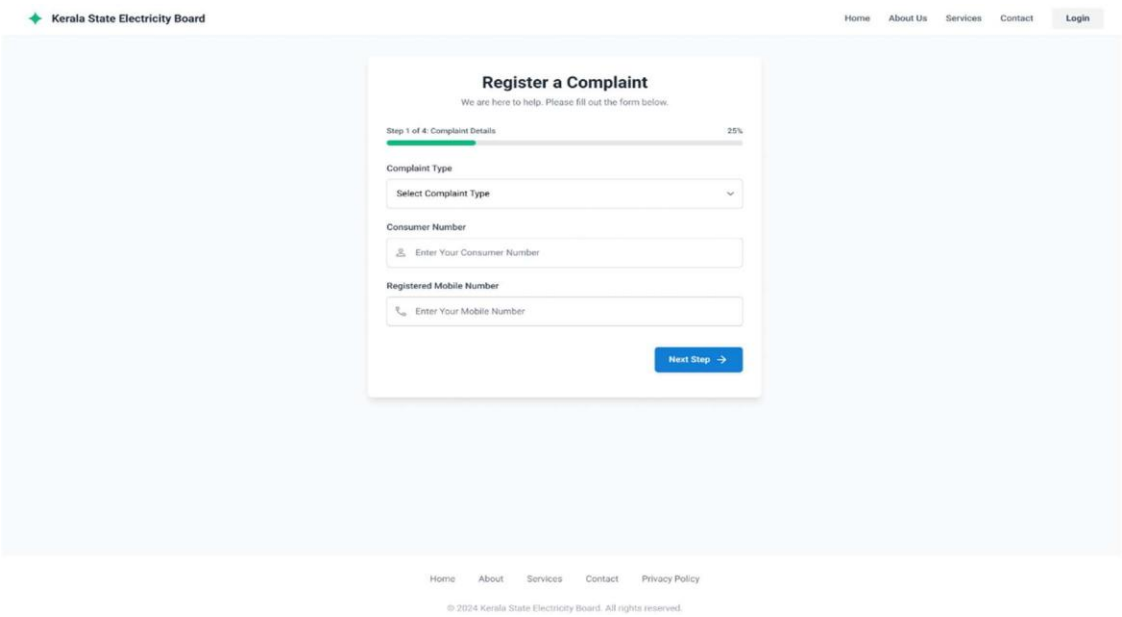
PROTOTYPE

Kerala State Electricity Board
KSEB
കേരളത്തിന്റെ ഊർജം
ENTER THE 13 DIGIT CONSUMER NUMBER
ENTER THE REGISTERED PHONE NUMBER
LOGIN
Sign In


Kerala State Electricity Board
Welcome to KSEB
Explore Services
Quick Actions
Pay Bill
Register Complaint
Track Complaint
Download Forms
View Tenders
Important Notices
Scheduled Power Outage


Kerala State Electricity Board
Powering Kerala's Future
Quick Services
Enter Consumer Number
Next
Explore Our Services
View Payment History
Manage Account
Consumption Insights
Latest Updates
New Renewable Energy Initiatives
Read More
Energy Efficiency Tips
About Us
Contact
Terms of Service
Privacy Policy


Customer Support
Green Initiatives
Energy Saving Tips
KSEB


Kerala State Electricity Board
Home
About Us
Services
Contact
Login
Register a Complaint
We are here to help. Please fill out the form below.
Step 1 of 4: Complaint Details
25%
Complaint Type
Select Complaint Type
Consumer Number
Enter Your Consumer Number
Registered Mobile Number
Enter Your Mobile Number
Next Step
Home
About
Services
Contact
Privacy Policy
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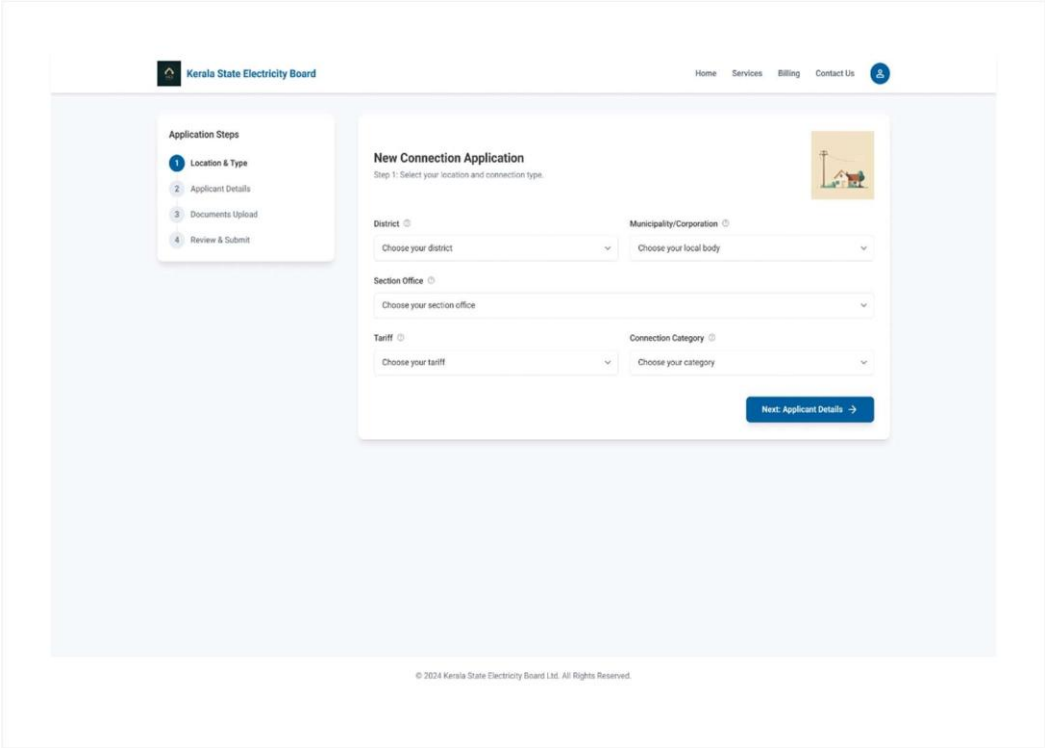
Kerala State Electricity Board
Home
Services
Billing
Contact Us
Application Steps
1 Location & Type
2 Applicant Details
3 Documents Upload
4 Review & Submit
New Connection Application
Step 1: Select your location and connection type.
District
Choose your district
Municipality/Corporation
Choose your local body
Section Office
Choose your section office
Tariff
Choose your tariff
Connection Category
Choose your category
Next: Applicant Details
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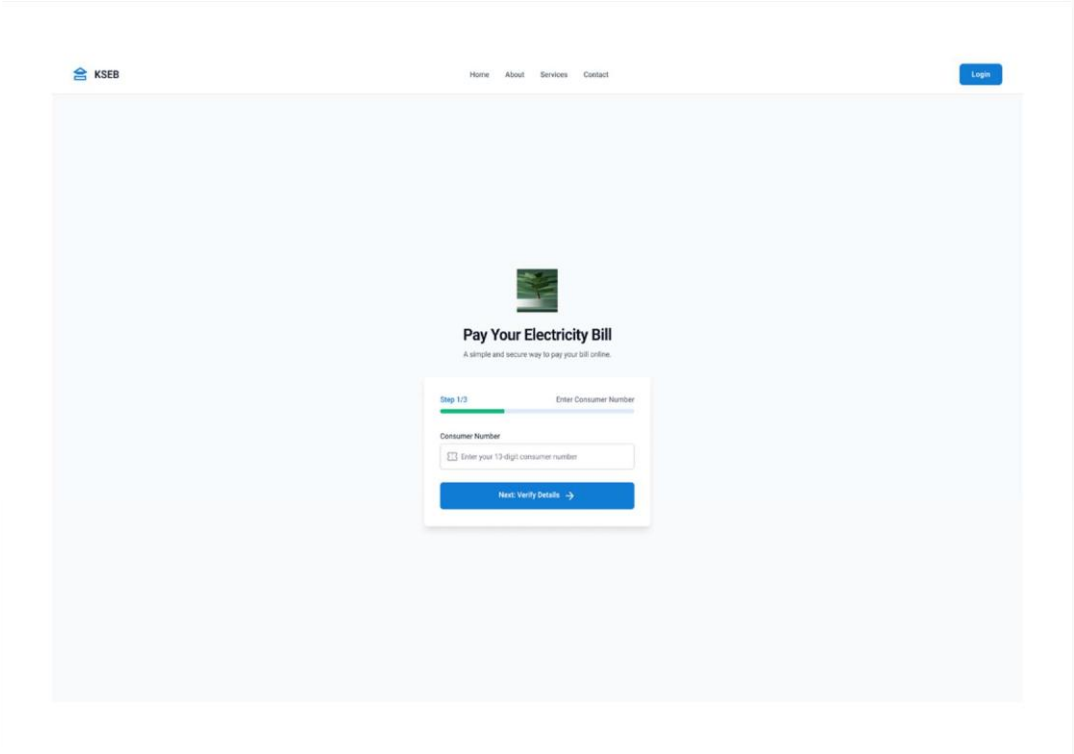
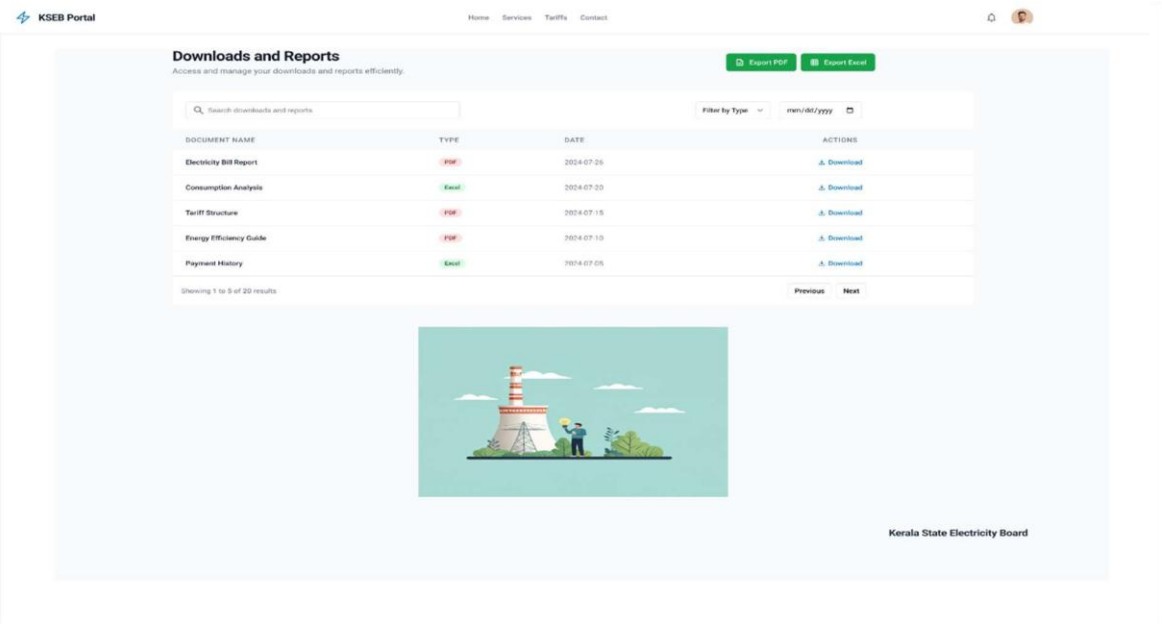
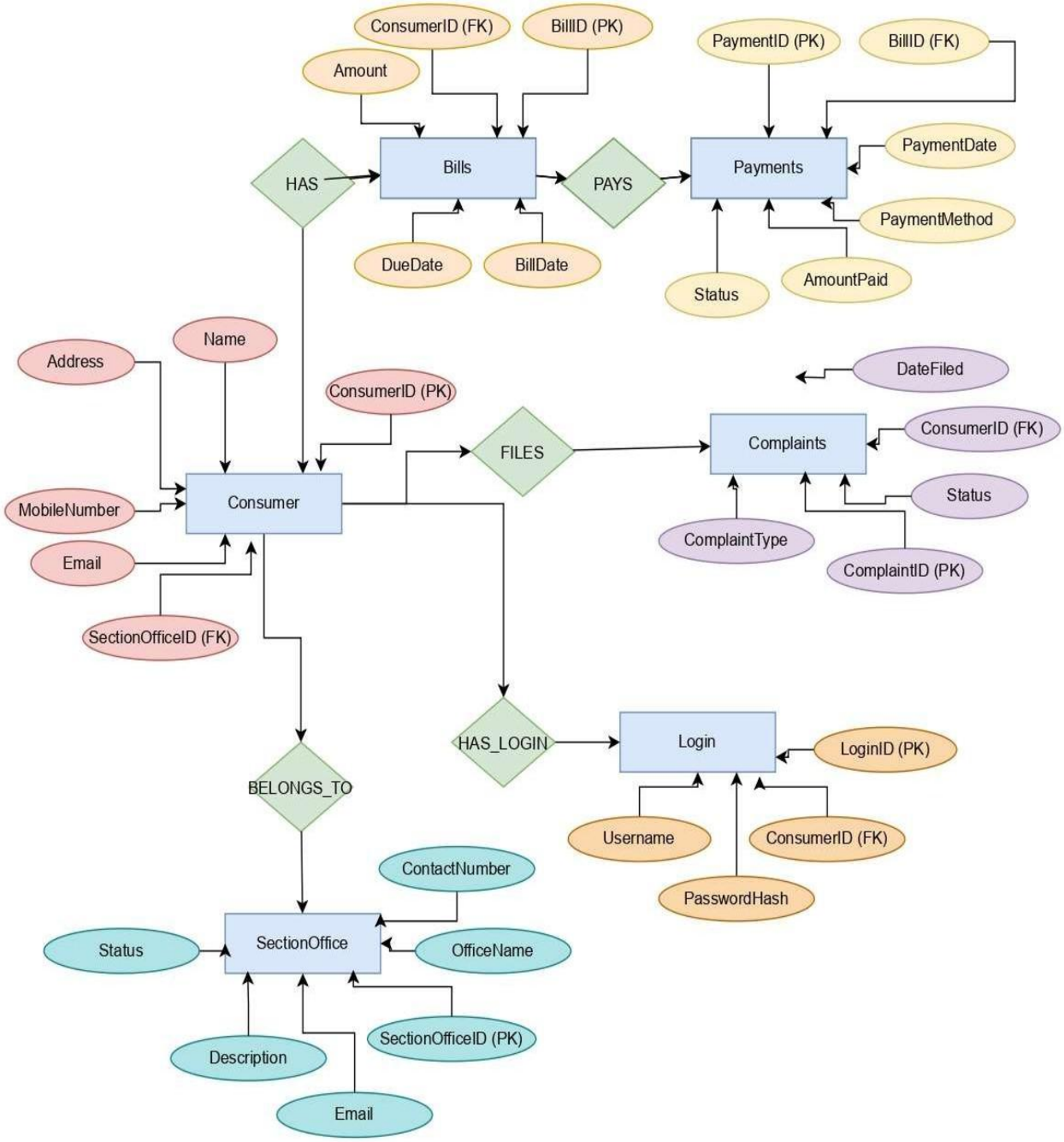

KSEB
Home
About
Services
Contact
Login
Pay Your Electricity Bill
A simple and secure way to pay your bill online.
Step 1/3
Enter Consumer Number
Consumer Number
Enter your 13-digit consumer number
Next: Verify Details


KSEB Portal
Home
Services
Tariffs
Contact
Downloads and Reports
Access and manage your downloads and reports efficiently.
Export PDF
Export Excel
Search downloads and reports
Filter by Type
mm/dd/yyyy
DOCUMENT NAME
TYPE
DATE
ACTIONS
Electricity Bill Report
PDF
2024-07-26
Download
Consumption Analysis
Excel
2024-07-20
Download
Tariff Structure
PDF
2024-07-15
Download
Energy Efficiency Guide
PDF
2024-07-10
Download
Payment History
Excel
2024-07-05
Download
Showing 1 to 5 of 20 results
Previous
Next
Kerala State Electricity Board

ER-Diagram


ConsumerID (FK)
BillID (PK)
Amount
PaymentID (PK)
BillID (FK)
HAS
Bills
PAYS
Payments
PaymentDate
PaymentMethod
DueDate
BillDate
Status
AmountPaid
Address
Name
DateFiled
ConsumerID (PK)
FILES
Complaints
ConsumerID (FK)
MobileNumber
Consumer
Status
ComplaintType
Email
ComplaintID (PK)
SectionOfficeID (FK)
HAS_LOGIN
Login
LoginID (PK)
BELONGS_TO
Username
ConsumerID (FK)
ContactNumber
PasswordHash
Status
SectionOffice
OfficeName
SectionOfficeID (PK)
Description
Email

Relational Schema

Entity	Attributes	Primary Key (PK)	Foreign Keys (FK)
Consumer	ConsumerID, Name, Address, MobileNumber, Email, SectionOfficeID	ConsumerID	SectionOfficeID → SectionOffice
Bills	BillID, ConsumerID, BillDate, DueDate, Amount	BillID	ConsumerID → Consumer
Payments	PaymentID, BillID, PaymentDate, PaymentMethod, AmountPaid, Status	PaymentID	BillID → Bills, ConsumerID → Consumer
Complaints	ComplaintID, ConsumerID, ComplaintType, DateFiled, Status	ComplaintID	ConsumerID → Consumer, SectionOfficeID → SectionOffice
SectionOffice	SectionOfficeID, OfficeName, ContactNumber, Email, Description, Status	SectionOfficeID	
Login	LoginID, ConsumerID, Username, PasswordHash	LoginID	ConsumerID → Consumer

SQL Queries for Consumer Table

The Kerala State Electricity Board (KSEB) wants to digitize its consumer management system. You are tasked with designing and managing the 'Consumer' table with operations covering DDL, DML, SELECT queries, aggregate functions, ordering, and set operations. Below are the scenario-based tasks and solutions.

1. DDL – Create and Alter

Create Consumer table

```
CREATE TABLE Consumer (
   ConsumerID INT PRIMARY KEY,
   Name VARCHAR(100) NOT NULL,
   Address VARCHAR(200),
   MobileNumber VARCHAR(15),
   Email VARCHAR(100),
   SectionOfficeID INT,
   ConsumerType VARCHAR(50) CHECK (ConsumerType IN ('Domestic',
'Commercial', 'Industrial')),
   ConnectionDate DATE,
   Status VARCHAR(20) CHECK (Status IN ('Active', 'Inactive'))
);
```

Add AadhaarNumber column later

ALTER TABLE Consumer

```
ADD AadhaarNumber VARCHAR(12) UNIQUE;
```

2. DML – Insert, Update, Delete

Insert 3 consumers

```
INSERT INTO Consumer
(ConsumerID, Name, Address, MobileNumber, Email, SectionOfficeID, ConsumerType,
ConnectionDate, Status, AadhaarNumber)
VALUES
(1, 'Anil Kumar', 'Trivandrum', '9876543210', 'anil.kumar@email.com', 101, 'Domestic',
'2022-01-15', 'Active', '123456789012'),
(2, 'Binu Joseph', 'Ernakulam', '9876501234', 'binu.joseph@email.com', 102,
```

'Commercial', '2021-11-05', 'Active', '234567890123'),
(3, 'Asha Varma', 'Trivandrum', '9876001111', 'asha.varma@email.com', 101, 'Industrial', '2023-03-20', 'Active', '345678901234');

Update: Change ConsumerID = 1 to Inactive

UPDATE Consumer
SET Status = 'Inactive'
WHERE ConsumerID = 1;

Delete: Remove ConsumerID = 2
DELETE FROM Consumer
WHERE ConsumerID = 2;

3. SELECT with WHERE and LIKE

Active consumers in Trivandrum whose names start with 'A'

```
SELECT ConsumerID, Name, Address, Status
FROM Consumer
WHERE Status = 'Active'
  AND Address = 'Trivandrum'
  AND Name LIKE 'A%';
```

4. Aggregate with GROUP BY and HAVING

Count active consumers per SectionOffice, only if more than 2

```
SELECT SectionOfficeID, COUNT(*) AS ActiveConsumers
FROM Consumer
WHERE Status = 'Active'
GROUP BY SectionOfficeID
HAVING COUNT(*) > 2;
```

5. ORDER BY

List consumers in order of connection date (earliest first)

```
SELECT ConsumerID, Name, ConnectionDate, Status
FROM Consumer
ORDER BY ConnectionDate ASC;
```

7. Set Operation

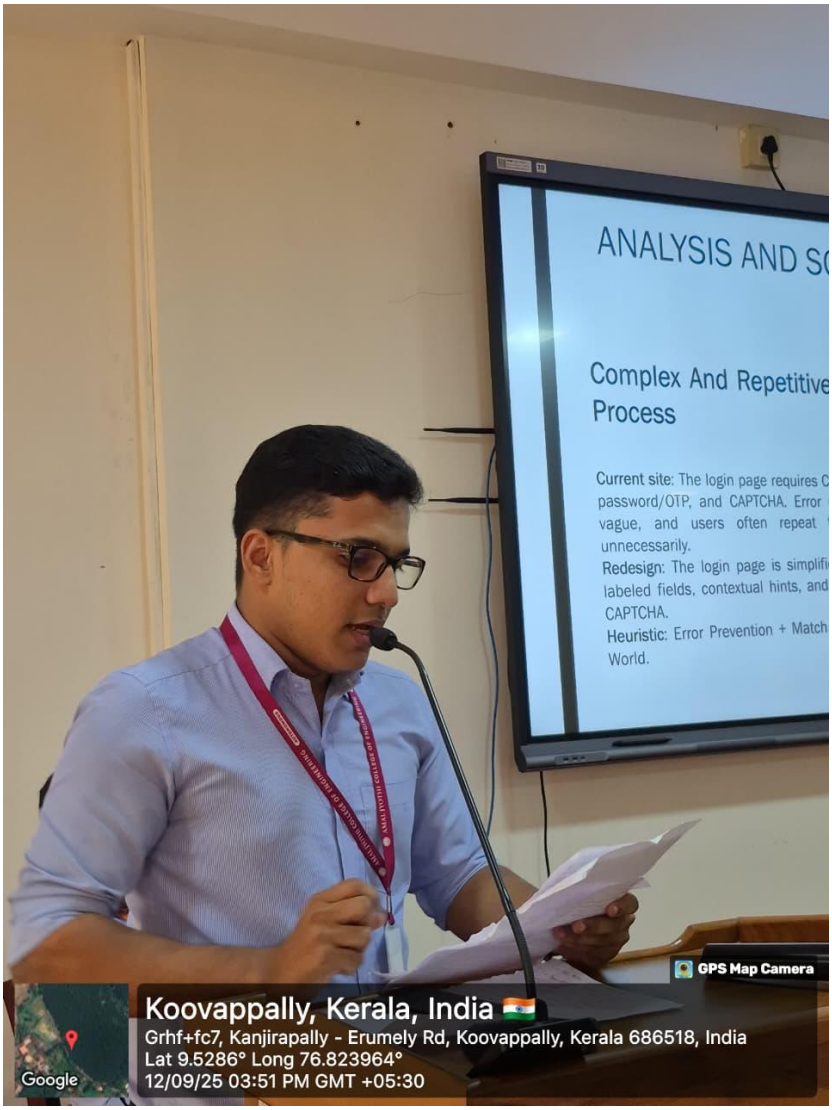
KSEB wants a list of all consumers from SectionOfficeID 101 and 102, without duplicates.

```
SELECT ConsumerID, Name, SectionOfficeID
FROM Consumer
WHERE SectionOfficeID = 101
UNION
SELECT ConsumerID, Name, SectionOfficeID
FROM Consumer
WHERE SectionOfficeID = 102;
```

Find all consumers who are NOT in SectionOfficeID 101.

```
SELECT ConsumerID, Name
FROM Consumer
EXCEPT
SELECT ConsumerID, Name
FROM Consumer
WHERE SectionOfficeID = 101;
```

GEO-TAGGED PHOTOS



NEW CHANGES BROUGHT TO THE WEBSITE

Report on Redesigned KSEB Quick Pay Portal

1. Introduction

1. The Kerala State Electricity Board (KSEB) portal is an essential platform used by millions of consumers for paying bills, registering complaints, and applying for new electricity connections. However, the existing website has several usability issues that make it difficult for consumers to access key services easily.
   This report highlights the redesigned KSEB Payment Portal, focusing on its new features, user-friendly design, and performance improvements compared to the old portal.

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2. Analysis of the Existing Portal

2. **Old Website:** https://wss.kseb.in/selfservices/quickpay

Existing Issues

3. **Outdated Layout:** The site looks old-fashioned and lacks visual appeal or clear organization.
4. **Complicated Navigation:** Users must go through multiple pages or menus to find what they need.
5. **Confusing Login Process:** Repeated CAPTCHA verification and unclear error messages frustrate users.
6. **Unclear Payment and Complaint Options:** Forms are not intuitive and often lead to incorrect submissions.
7. **Difficult Access to Receipts:** Users need to log in multiple times or check emails to access receipts.
8. **No Guidance for Multi-Step Tasks:** Processes like applying for a new connection have no progress indication, leaving users uncertain about the next step.
9. Overall, the current system feels slow, repetitive, and not user-friendly — especially for first-time or elderly users.

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3. Objectives of the Redesign

10. The redesigned portal aims to:

- Simplify the overall user experience.
- Enable quick and secure access to major services.
- Improve accessibility across all devices (desktop and mobile).
- Enhance visual design with clear sections and icons.
- Streamline payment, complaint, and connection processes.
- Provide direct access to receipts and application progress.
- Create a more modern and efficient digital platform for KSEB consumers.

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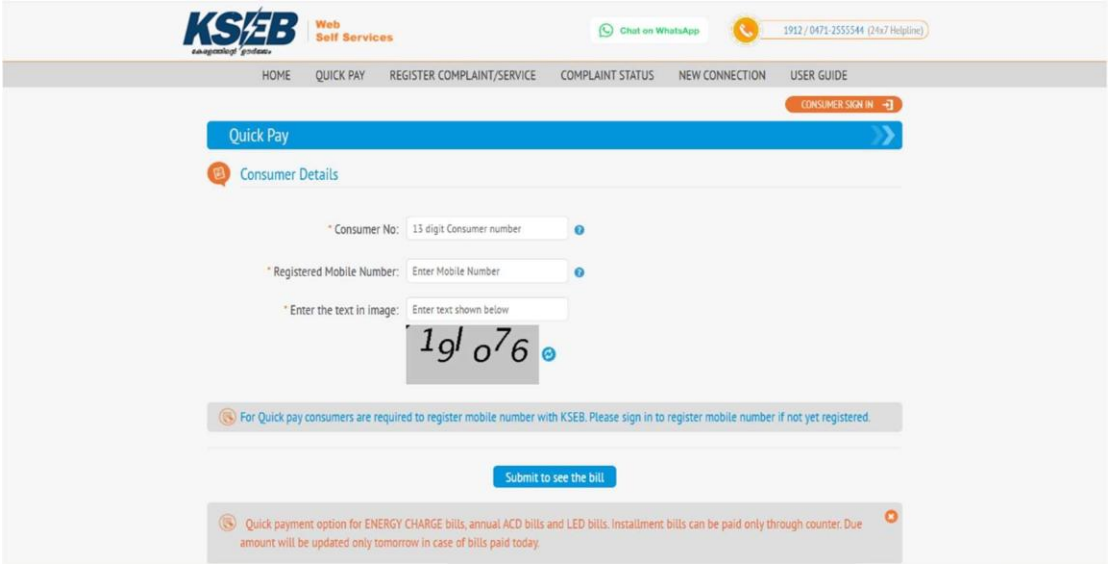
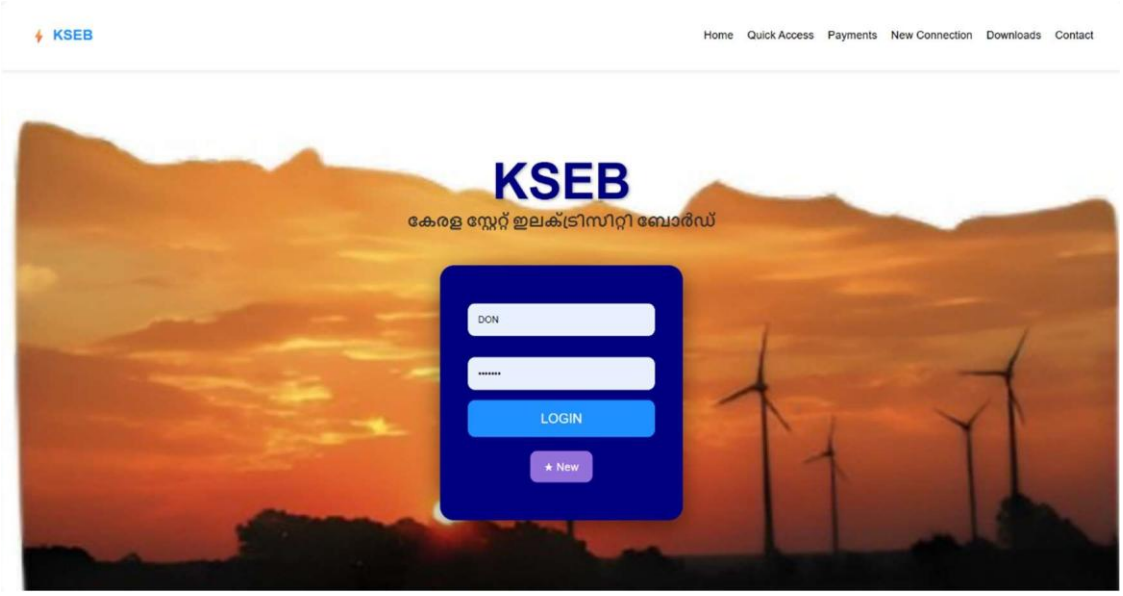
4. Features of the Redesigned Portal

a) New Login Page

11. **Old:** The old login page required the Consumer ID, password/OTP, and a CAPTCHA for every session. Error messages were unclear, causing users to re-enter data multiple times.



New: The redesigned login page has a clean layout with clearly labeled fields and simple instructions. CAPTCHA has been minimized, and the process is much faster.



Explanation: This design helps users log in smoothly without unnecessary repetition and confusion. The page looks modern and works well on all devices.

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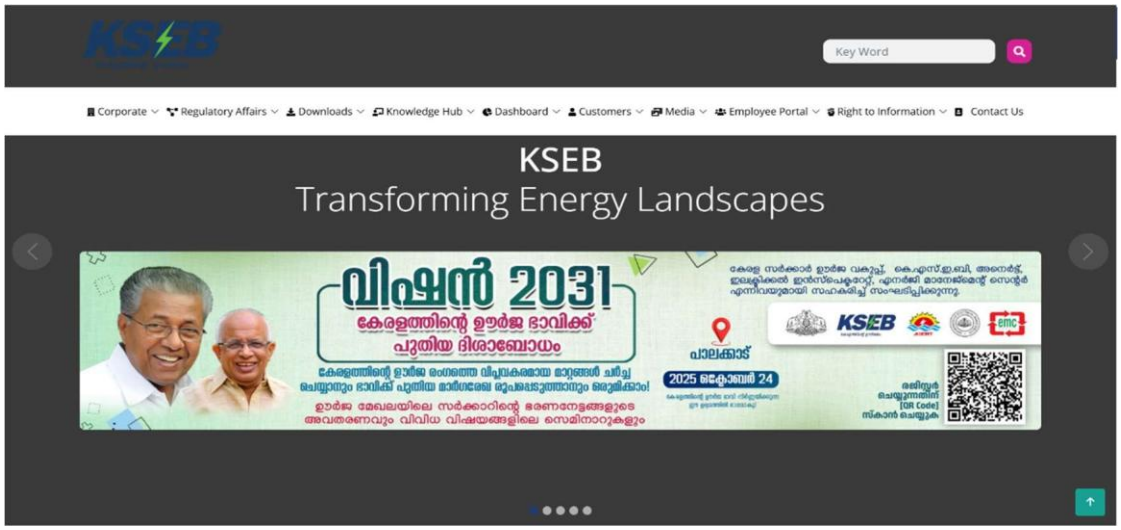
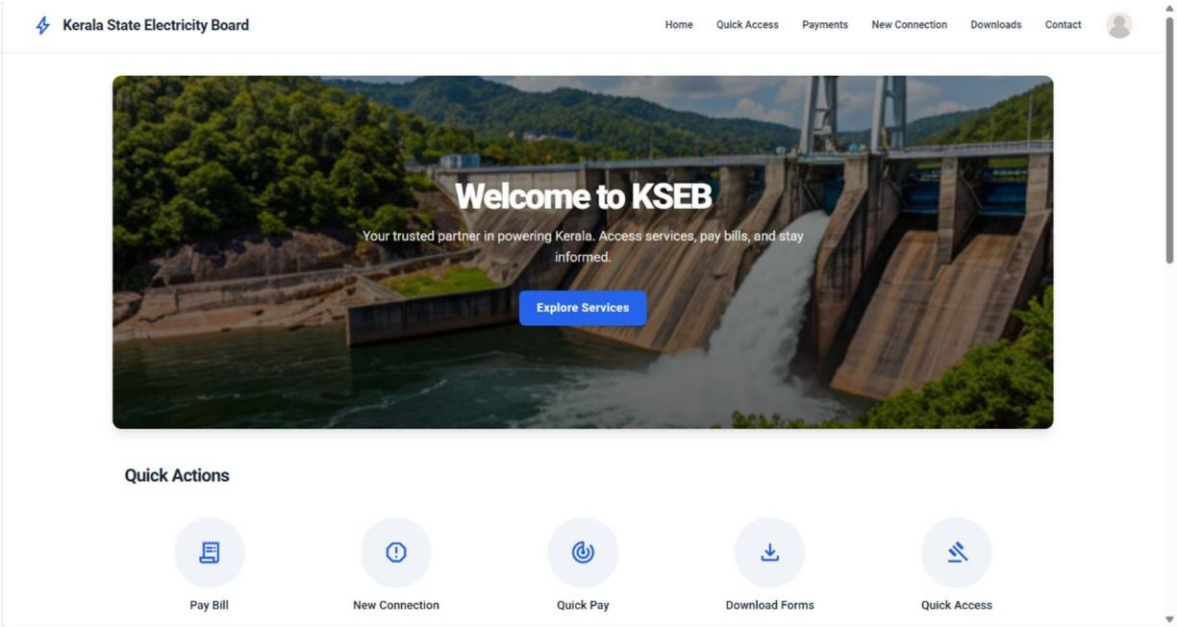
b) Easy Accessibility and Navigation

12. **Old:** Services like Quick Pay, Complaint, and New Connection were listed as plain text links that were hard to find.



New: The homepage now displays all key services such as *Bills*, *Quick Pay*, *Complaints*, *New Connection*, and *Downloads* as attractive cards with icons.



Explanation: Users can identify services easily without searching through menus. The interface is clean and organized, ensuring faster navigation.

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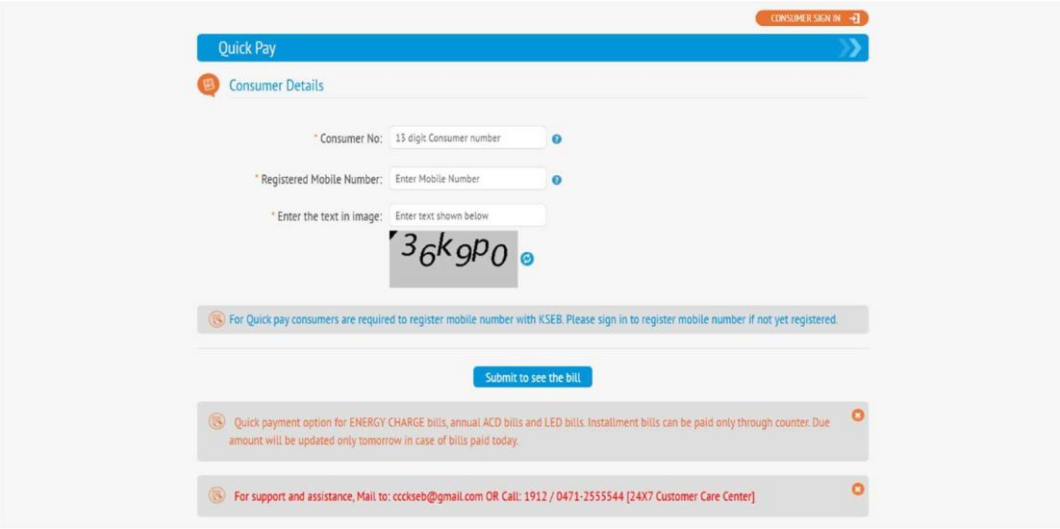
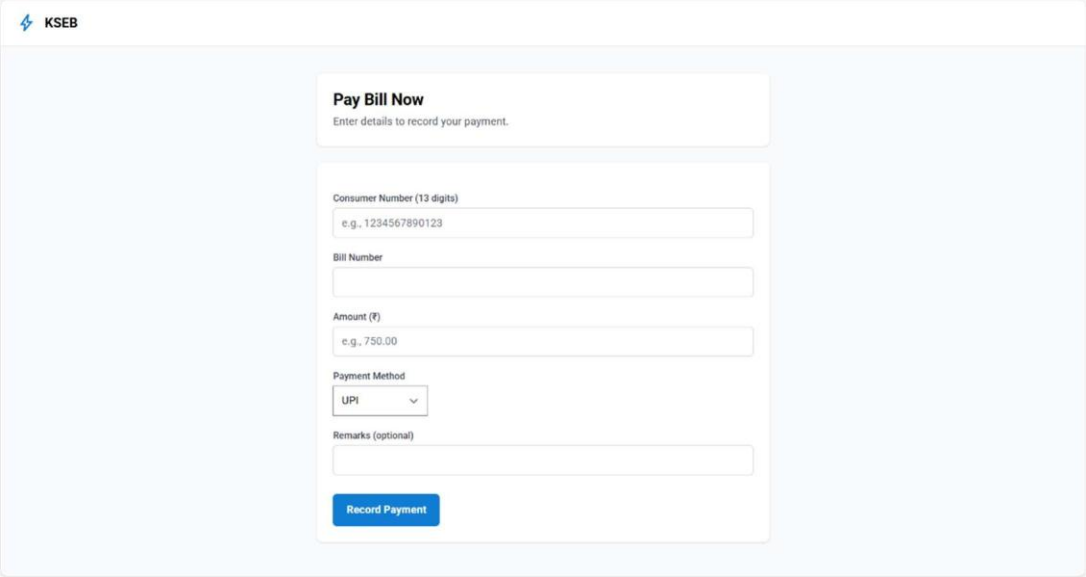
c) Quick Pay Improvement

13. **Old:** The Quick Pay form was plain and lacked instructions, often leading to incorrect or incomplete entries.



New: The redesigned Quick Pay section includes clear placeholders and tooltips explaining what each field means. Steps are reduced, and users are guided smoothly from input to payment.



Explanation: This improvement makes payment faster, easier, and more accurate, reducing errors during the process.

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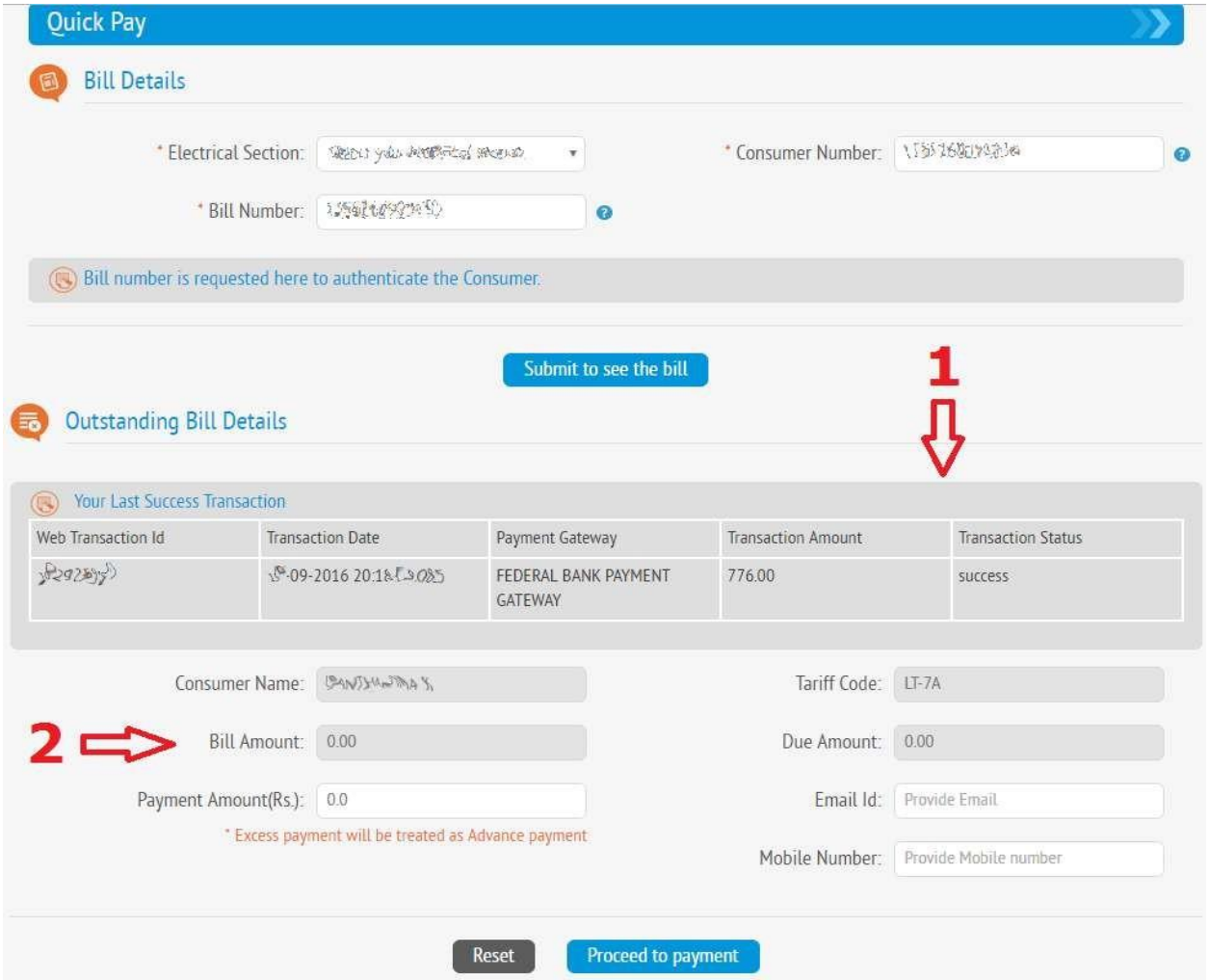
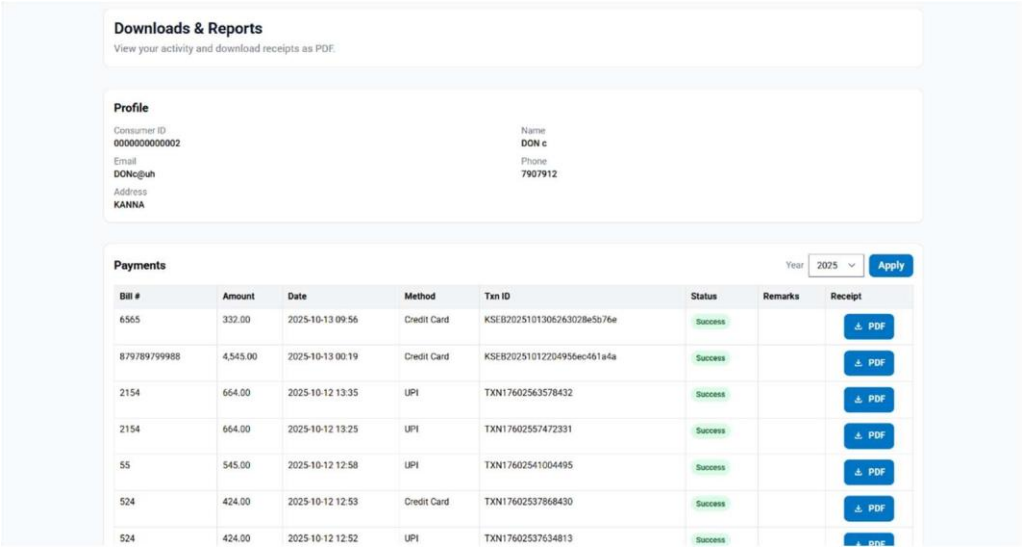
d) Easy Receipt Download

14. **Old:** Receipts and payment records were difficult to find. Users often had to check their emails or go through multiple logins to retrieve them.



New: The redesigned version includes a dedicated **Downloads** section that allows users to view and download receipts directly from their account.



Explanation: This ensures that users can access their records quickly and conveniently without extra steps.

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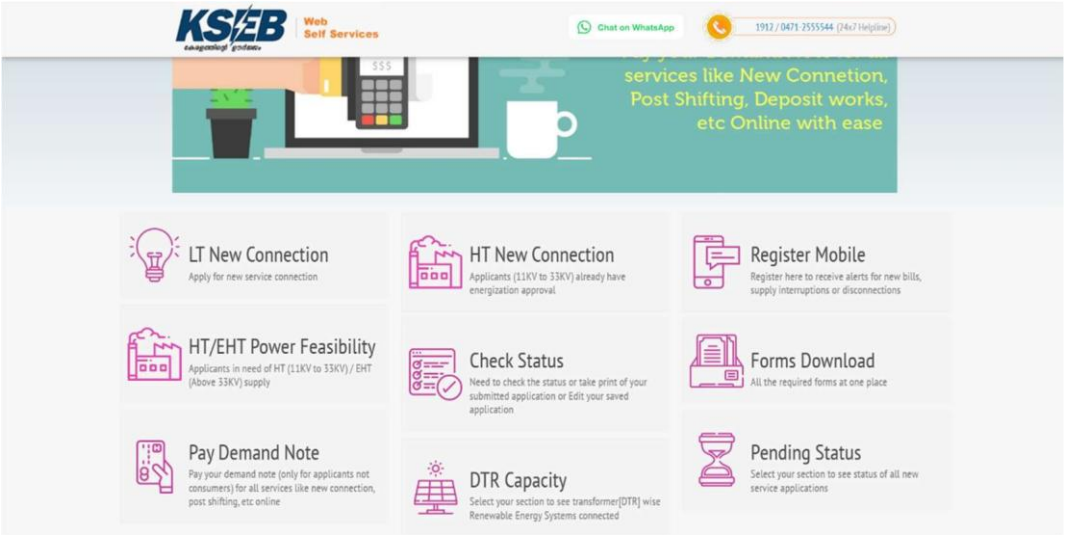
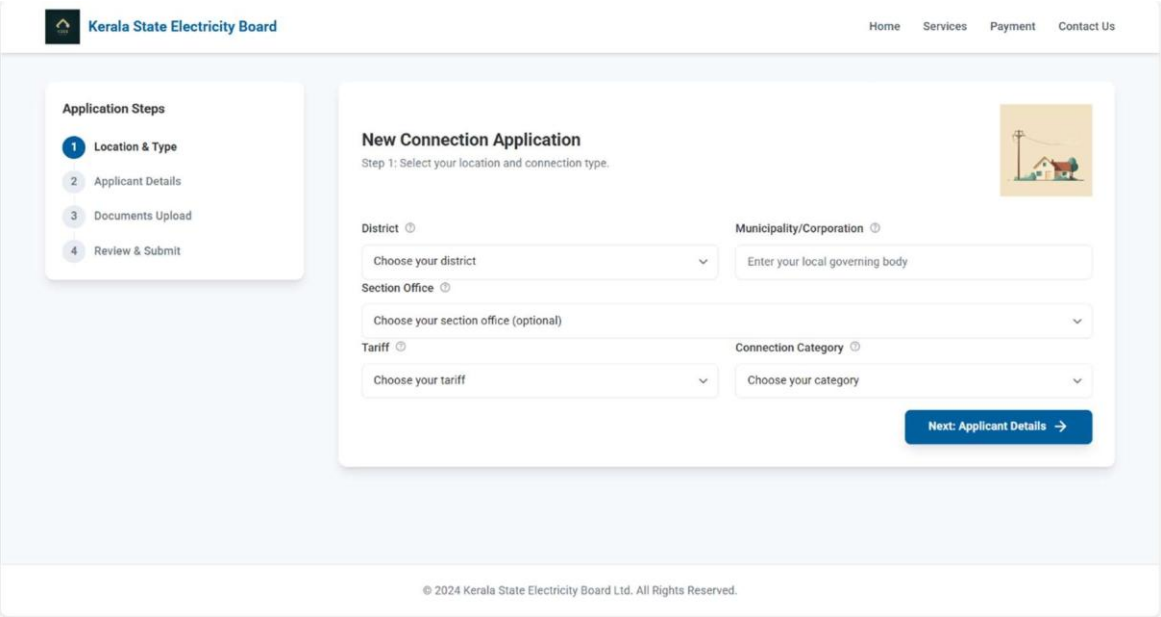
e) Progress Indicator in New Connection Section

15. **Old:** The process of applying for a new connection was divided across multiple pages with no indication of how many steps were remaining.



New: A new **progress bar** has been added to the New Connection section, showing each stage — from application submission to verification and payment confirmation.



Explanation: This helps users understand where they are in the process and reduces confusion about what to do next.

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5. Additional Enhancements

- **Modern Design Theme:** A more professional layout with better color contrast for readability.
- **Mobile-Friendly Interface:** Fully responsive design that adjusts to phones, tablets, and desktops.
- **Search Option:** A built-in search feature allows quick access to any service.
- **Improved Dashboard:** Displays bill status, payment history, and account details clearly.
- **Database Integration:** Backend tables and relationships are structured efficiently to handle data securely.
- **Prototype Design:** The portal has been visually prototyped using Figma for testing and design validation.

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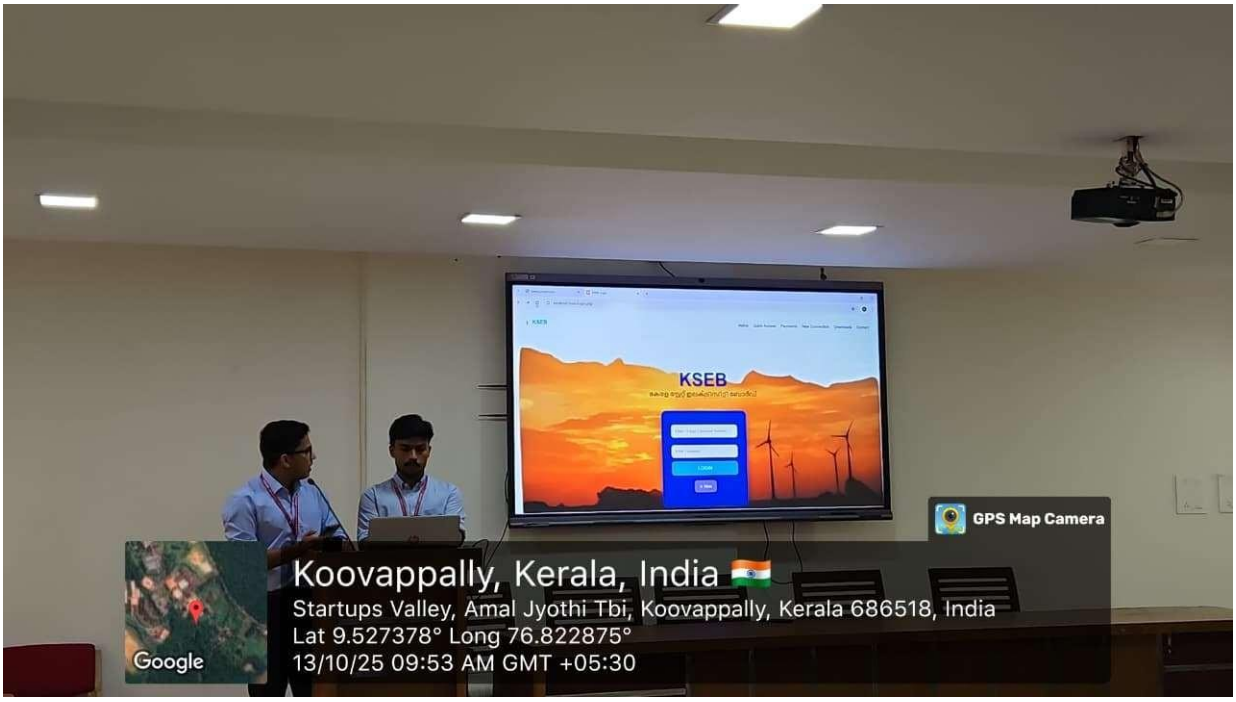
6. Comparison Between Old and New Portal

Feature	Old KSEB Portal	Redesigned KSEB Portal
Login Process	Repetitive CAPTCHA, unclear prompts	Simple, clean, and faster login
Navigation	Text-based menus	Icon-based service cards
Quick Pay	Plain form, no guidance	Clear placeholders and tooltips
Complaint Registration	Confusing and hidden	Dropdown-based, structured form
Receipt Download	Scattered and email-based	Centralized and easily accessible
New Connection	Multiple unclear steps	Step-by-step progress indicator
Mobile Accessibility	Limited	Fully responsive and mobile-ready

7. Conclusion

16. The redesigned **KSEB Quick Pay Portal** successfully transforms the traditional user experience into a modern, fast, and accessible digital service.
    Users can now log in quickly, make payments effortlessly, download receipts instantly, and track new connection requests with clear progress visibility.
    This new design not only improves usability and accessibility but also aligns with the goal of digital transformation for Kerala's public service infrastructure.

WEBSITE LINK:
https://youtu.be/xS7nmiQv9pg?si=VRDLEwZ1pqGbK88x

GEOTAGGED -PHOTOS